A **Conditional FP-Tree (Conditional Frequent Pattern Tree)** is a substructure of an **FP-Tree (Frequent Pattern Tree)** used in the **FP-Growth algorithm** for mining frequent itemsets efficiently. It helps in recursive pattern mining without generating candidate itemsets explicitly, unlike the Apriori algorithm.

How Conditional FP-Tree Works

1. **Build the FP-Tree:**
   - Construct the FP-Tree from the transactional dataset.
   - Store items in descending order of frequency.
2. **Mine Frequent Itemsets Using Conditional FP-Trees:**
   - Start from the least frequent item in the FP-Tree (called the **conditional base pattern**).
   - Extract the **conditional pattern base**, which includes all transactions (prefix paths) where the item appears.
   - Construct a **conditional FP-Tree** from these prefix paths.
   - Recursively mine the conditional FP-Tree to find frequent itemsets.

Example

Transaction Dataset

Transaction	Items (sorted by frequency)
T1	A, B, C, D
T2	B, C, E
T3	A, B, C, E
T4	A, C, D, E
T5	B, C, D

Step 1: FP-Tree Construction

The FP-Tree is constructed based on item frequencies, grouping common prefixes.

Step 2: Conditional FP-Tree for Item "E"

- The **conditional pattern base** for E includes transactions contributing to its occurrence.
- The **conditional FP-Tree** for E is created using only these transactions.
- Frequent itemsets are recursively mined from this tree.

Benefits of Conditional FP-Tree

- **Avoids candidate generation**, unlike Apriori.
- **Efficiently handles large datasets** by recursively reducing the problem size.
- **Compresses frequent patterns** for faster mining.